USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting

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Abstract—A visual localizer should tell a robot two things: where the camera is, and how much to trust that answer. Trust is essential for safety-critical platforms such as indoor aerial robots, which fuse every pose measurement into a Bayesian state estimator. Recent semantic 3D Gaussian Splatting (3DGS) localizers are highly accurate. However, they return a single pose with no measure of confidence. Uncertainty-aware mapping methods quantify the reliability of every map element. However, they do not localize a query image. We close this gap with USGLoc, an evidential localization framework built on one principle: the map's own uncertainty should drive every stage of localization and survive to the output. USGLoc represents the scene as an evidential semantic Gaussian map. Each primitive carries a Dempster–Shafer belief over open-vocabulary semantics and an explicit vacuity (ignorance) mass. The query image is encoded into the same evidential space. Vacuity then acts as one continuous signal throughout the pipeline. It gates 2D–3D semantic matching, which turns retrieval ambiguity into a weighted pose mixture. It is alpha-blended into a per-pixel weight, so rendering-based refinement discounts moved and ambiguous content. A tempered Laplace approximation then converts each refined hypothesis into a covariance. The result is a calibrated Gaussian-mixture pose posterior on $SE(3)$, which condenses to a single Gaussian for direct use in a Bayesian filter. [TODO: one-sentence headline result]. Beyond accuracy, we evaluate calibration with coverage, sparsification error, and negative log-likelihood. These metrics are largely absent from prior localization work. They show that the predicted uncertainty is both calibrated and informative.

Index Terms—Localization, RGB-D perception, semantic scene understanding, uncertainty quantification, deep learning for visual perception.

I. Introduction

VISUAL localization estimates the 6-DoF camera pose of a query image inside a previously mapped scene. It is a core capability for indoor mobile robots and micro aerial vehicles (MAVs) [1]–[3]. On these platforms the pose is rarely consumed on its own. It enters a recursive Bayesian estimator, such as an EKF or UKF, and is fused with inertial and other cues [4]. This fusion is only as sound as the uncertainty attached to each measurement. A pose without a covariance forces the estimator to assume a fixed noise model. A fixed model cannot reflect scene ambiguity, occlusion, dynamics, or texture sparsity. These are exactly the conditions under which localization degrades. An indoor localizer should therefore output a calibrated distribution over pose, not a point.

Classical localization is dominated by feature-based and regression-based methods. Feature-based pipelines match 2D keypoints to a 3D model and solve Perspective-n-Point with RANSAC [5]. Absolute pose regression (APR) [6], [7] and scene coordinate regression (SCR) [8], [9] learn the mapping end-to-end. 3D Gaussian Splatting (3DGS) [1] then brought high-fidelity novel-view synthesis at real-time rates. It has become an attractive map representation, and a family of 3DGS localizers now refines pose by inverting or matching against the rendered map [10]–[13]. Semantics improve robustness further. SGLoc [2] retrieves a coarse pose by aligning 2D and 3DGS scene-level semantic descriptors. Chu and Kim [3] use a language model to select "stable" objects for a robust submap. These methods are accurate, but they share two limitations. First, they emit a single pose and say nothing about its reliability. Second, they handle scene dynamics implicitly or through hard, discrete object selection. Discrete selection cannot express graded confidence.

A separate line of work quantifies uncertainty rigorously, but it stops short of a calibrated pose distribution over a 3DGS map. Wu and Campbell [14] learn a measurement-error distribution for image-retrieval localization. They show that the error is non-Gaussian under hard conditions and is best modeled by a Gaussian mixture. Their setting is purely 2D, with no 3D map and no rendering. Particle-based localizers over radiance fields [15] maintain a nonparametric pose belief. They report only its MAP point, without calibration. Most recently, UGS-Loc [16] guides 3DGS pose refinement with Fisher-information uncertainty maps and Monte Carlo pose sampling. It uses uncertainty only internally and also emits a single point estimate. Evidential learning has reached scene-coordinate regression [17], but without a metric map at test time. On the mapping side, evidential semantic mapping [18], [19] propagates per-observation uncertainty with Evidential Deep Learning (EDL) [20] and Dempster–Shafer Theory (DST) [21]. E^2-BKI [19] adds anisotropic Gaussian primitives and geometry-aligned kernels. Rendering-side uncertainty for radiance fields is also well studied [22]. These methods produce trustworthy

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maps. Their uncertainty, however, never reaches the pose estimate that a robot consumes. To our knowledge, no existing method outputs a calibrated pose distribution from a Gaussian-splatting map (Fig. 2).

USGLoc fills that cell. Its design follows one principle: the map's own uncertainty should drive every stage of localization and survive to the output. The scene is represented as an evidential semantic Gaussian map. Each object-level primitive stores geometry, an open-vocabulary CLIP [23] embedding, and a DST belief with an explicit vacuity mass. Vacuity is the evidential measure of ignorance. It is created once, when observations are encoded, and it is never discarded. The query image is segmented [24] and encoded by the same evidential head. Query and map confidences therefore live on one scale. Every later stage is then derived as a consumer of this single signal. In the coarse stage, vacuity gates 2D–3D semantic matching, and retrieval ambiguity becomes an explicit weighted pose mixture. In the fine stage, vacuity is alpha-blended through the rasterizer exactly like color. The result is a per-pixel trust map, and moved or ambiguous content loses its vote on the pose. At the optimum, a tempered Laplace approximation turns the weighted objective into a per-hypothesis covariance. The output is a Gaussian-mixture posterior on $SE(3)$, condensed to a single Gaussian for a downstream filter. No stage requires dynamic-object heuristics or a language model. Our contributions are:

- Evidential localization on a 3DGS map. We introduce the first localizer that outputs a calibrated $SE(3)$ pose distribution from a Gaussian-splatting map. Prior 3DGS localizers ignore uncertainty [11]–[13], consume it internally and return a point [16], or maintain uncalibrated beliefs in non-splatting settings [15].
- One propagated uncertainty channel. Every stage consumes the same evidential state. Conflict-aware DST fusion builds the map. Vacuity gates semantic matching and turns ambiguity into a weighted pose mixture, lifting the non-Gaussian-error insight of [14] from 2D retrieval to 6-DoF. Vacuity rendered through the standard alpha-blending kernel weights the refinement per pixel, replacing discrete robust-object selection [3] with graded, evidence-driven trust.
- Calibrated covariance from rendering. A Laplace approximation at the refinement optimum converts the weighted photometric objective into a per-hypothesis covariance at negligible cost. A single temperature, fitted on a held-out split, makes it honest.
- A calibration-centric protocol. We evaluate coverage, AUSE, NLL, and ECE on standard and semi-dynamic indoor benchmarks. [TODO: summary of empirical findings].

II. Related Work

A. Visual Localization

APR methods directly regress pose from an image [6], [7]. They are fast and storage-light. However, they tie accuracy to the training distribution. They also lack explicit geometric reasoning. SCR methods regress 2D–3D correspondences and recover pose by PnP+RANSAC [8], [9]. Geometry improves their accuracy. However, they generalize poorly across scenes and degrade in cluttered indoor layouts. Recent extensions add per-coordinate uncertainty to prefilter correspondences [25]. Feature-based pipelines [5] remain strong. However, they rely on low-level features that are sensitive to lighting and texture. Some of these methods do estimate uncertainty. In every case, that uncertainty is consumed internally. None reports a calibrated pose covariance.

B. Rendering-Based Localization (NeRF and 3DGS)

NeRF-based methods synthesize novel views to refine pose. They suffer from slow rendering and lighting sensitivity. Particle-based variants [15] maintain a nonparametric pose belief by render-and-compare. However, they report only its MAP point. 3DGS [1] enables fast, high-quality rendering and a compact map. Localization against 3DGS maps has therefore expanded rapidly. iComMa [10] inverts the rendering process by gradient-based optimization. GS-CPR [12], GSplatLoc [11], STDLoc [13], and the concurrent SplatHLoc [26] establish increasingly accurate matching and refinement against the rendered map. UGS-Loc [16] is the most related method. It refines an SCR prior on a 3DGS map with Monte Carlo pose particles. The particles are weighted by matcher confidence and by a rendered Fisher-information uncertainty map. The same uncertainty map biases PnP-RANSAC sampling toward low-uncertainty correspondences. Its uncertainty is post-hoc and purely geometric. It is derived from the photometric objective of a fixed map. It is also consumed internally: the output is the best particle, a single pose without covariance, calibration, or semantics. Per-correspondence uncertainty gating likewise appears in scene-coordinate regression [25]. In both cases the distribution dies before the output. SGLoc [2] adds semantic 2D-to-3DGS retrieval for prior-free coarse pose. Chu and Kim [3] augment a 3DGS map with semantics and an LLM-selected robust-object mask for dynamic indoor scenes. This mask performs discrete gating and has documented failure cases. Those failures motivate our continuous, evidence-driven weighting. These works substantiate our pipeline choices (semantic retrieval, coarse-to-fine, rendering refinement). However, they uniformly output point poses. Their robustness to dynamics is implicit or relies on hard object selection. USGLoc retains their accuracy mechanisms. It adds propagated, graded uncertainty and a calibrated pose distribution.

C. Uncertainty in Localization and Semantic Mapping

Bayesian and ambiguity-aware pose regressors output uncertainty, but they are limited to static scenes with dense pose annotations. Evidential learning has been applied to scene-coordinate regression with NeRF-derived uncertainty [17]. That approach has no metric

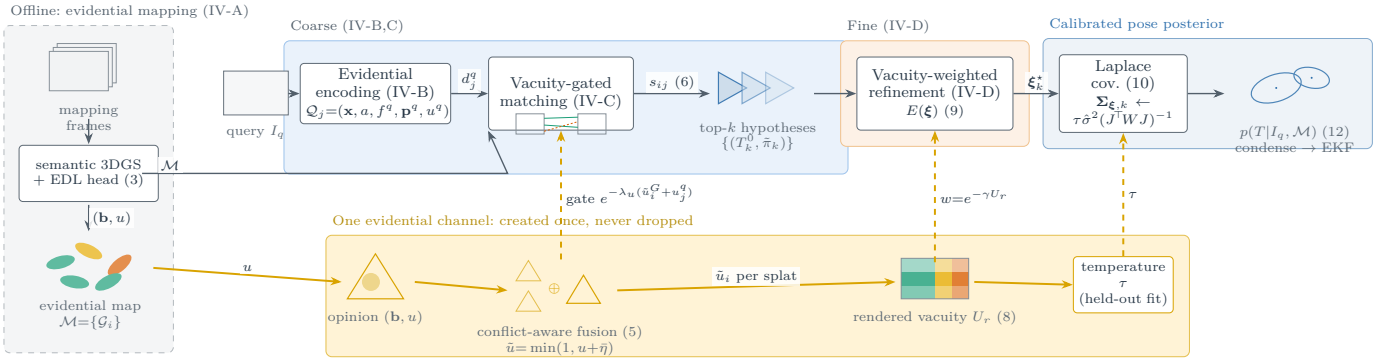


Fig. 1. Overview of USGLoc. Offline (left, grey): mapping frames train a semantic 3DGS map with an EDL head. Conflict-aware DST fusion turns per-observation opinions $(\mathbf{b}, u)$ into evidential primitives $\mathcal{G}_i = (\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, f_i, \mathbf{p}_i, \bar{u}_i)$. Online (top): the query is encoded by the same evidential head, matched against the map, and refined by differentiable rendering. The result is a calibrated Gaussian-mixture posterior $p(T \mid I_q, \mathcal{M})$ that condenses to a single Gaussian for an EKF. One evidential channel (bottom, gold): the same vacuity signal is injected at exactly three points. It gates matching, it weights the rendering residual per pixel, and it scales the Laplace covariance through the held-out temperature τ. Equation numbers in parentheses.

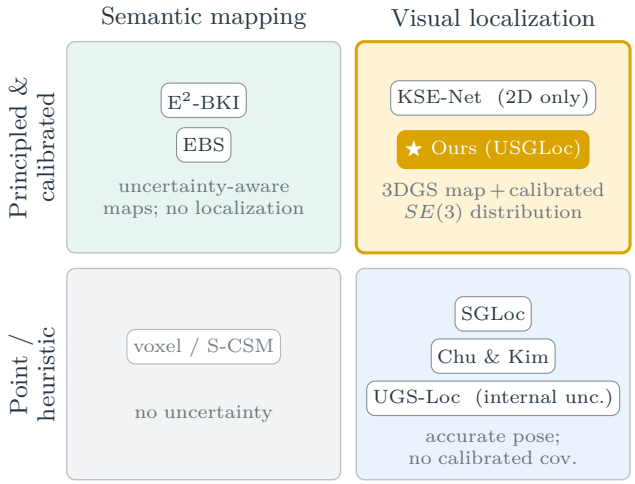


Fig. 2. Design-space positioning. Prior work covers uncertainty-aware semantic mapping (top-left) [18], [19]. It also covers accurate but point-estimate localization (bottom-right) [2], [3], [13]. Wu and Campbell [14] model a distribution, but only for 2D image retrieval. UGS-Loc [16] exploits uncertainty inside a 3DGS refiner, yet it still returns a point. USGLoc fills the open cell: a calibrated $SE(3)$ pose distribution from a 3DGS map.

map at query time. Wu and Campbell [14] learn a 2D measurement-error distribution from keypoint and semantic cues. They show the error is best modeled as a Gaussian mixture under adverse conditions. We adopt this distributional view. We realize it in 3D over $SE(3)$. In semantic mapping, EBS [18] and E^2-BKI [19] use EDL [20] and DST [21] to model semantic uncertainty. E^2-BKI represents the scene as anisotropic Gaussian primitives. Map- and rendering-side uncertainty for radiance fields is likewise active, for example through Fisher information [22]. In all of these, the uncertainty remains internal to mapping or tracking. We transplant the evidential machinery from mapping to localization. In mapping, it scores observations. In our localizer, it scores correspondences and pixels. This closes the gap between rigorous map-side uncertainty and the pose estimate consumed downstream.

III. Problem Formulation

Let $\mathcal{M} = \{\mathcal{G}_i\}_{i=1}^M$ be a pre-built semantic Gaussian map and I_q a query RGB(-D) image. Prior localizers seek a point estimate $\hat{T} = \arg\max_T s(I_q, T; \mathcal{M})$ for some scoring function s and a pose $T = [R \mid t] \in SE(3)$. We instead estimate the full posterior

$$p\big(T \,\big|\, I_q, \mathcal{M}\big), \qquad T \in SE(3), \tag{1}$$

and require it to be calibrated: a credibility-α region of (1) should contain the ground-truth pose with empirical frequency α. To make uncertainty first-class on both sides, we represent each modality as a set of evidential semantic instances. A reference primitive is

$$\mathcal{G}_i = \big(\boldsymbol{\mu}_i,\ \boldsymbol{\Sigma}_i,\ f_i,\ \mathbf{p}_i,\ u_i\big), \tag{2}$$

with mean $\boldsymbol{\mu}_i \in \mathbb{R}^3$, covariance $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$ (anisotropic geometry), CLIP embedding $f_i \in \mathbb{R}^d$, semantic probability $\mathbf{p}_i \in \Delta^{C-1}$ over C categories, and a scalar vacuity (uncertainty) mass $u_i \in [0, 1]$. A query instance $\mathcal{Q}_j = (\mathbf{x}_j, a_j, f_j^q, \mathbf{p}_j^q, u_j^q)$ stores its 2D centroid $\mathbf{x}_j$, area a_j, and the same semantic triplet. The remainder of the paper specifies each step. Sec. IV-A describes how $\mathcal{M}$ is built. Sec. IV-B describes how I_q is encoded. Secs. IV-C and IV-D describe how (1) is produced by uncertainty-aware matching and refinement.

IV. Method

Fig. 1 summarizes USGLoc. We first describe the evidential semantic representation shared by the map and the query. We then describe the two-stage coarse-to-fine localizer that turns this representation into a calibrated pose distribution.

A. Evidential Semantic Gaussian Map

Evidential semantics. Consider a semantic segmentation network with an evidential head [20]. Its non-negative outputs are interpreted as evidence $e = [e_1, \ldots, e_C]$ for a Dirichlet density with parameters $\alpha_c = e_c + 1$. With

$S = \sum_{c=1}^{C} \alpha_c$, the expected class probability, belief mass, and vacuity are

$$\hat{p}_c = \frac{\alpha_c}{S}, \qquad b_c = \frac{\alpha_c - 1}{S}, \qquad u = \frac{C}{S}, \tag{3}$$

satisfying $u + \sum_c b_c = 1$. Vacuity u is large when total evidence S is small, i.e., when the network is uncertain. This is the per-observation uncertainty we will propagate.

From Gaussians to evidential primitives. We build the map with an open-vocabulary semantic 3DGS representation that assigns an identity/semantic feature to each Gaussian [27]. Gaussians are clustered by spatial and semantic affinity into object-level primitives (as in [3]). For cluster $\mathcal{C}_i$, geometry is summarized by the empirical mean $\boldsymbol{\mu}_i$ and covariance $\boldsymbol{\Sigma}_i$ of member centroids. The CLIP embedding f_i is a vacuity-weighted average of member embeddings. The semantic belief is obtained by DST-fusing the member beliefs. For two belief–vacuity pairs $(\mathbf{b}_1, u_1), (\mathbf{b}_2, u_2)$, Dempster's rule gives [19]

$$b^c = \tfrac{1}{1-\eta}\big(b_1^c b_2^c + b_1^c u_2 + b_2^c u_1\big), \quad u = \tfrac{1}{1-\eta}\, u_1 u_2, \tag{4}$$

where $\eta = \sum_{x \neq y} b_1^x b_2^y$ measures conflict. Applying (4) pairwise over $\mathcal{C}_i$ yields $(\mathbf{b}_i, u_i)$. This pair is converted back to $\mathbf{p}_i$ via $p_i^c = b_i^c + u_i/C$. Fusing within a cluster suppresses isolated mispredictions. One subtlety is essential for dynamic scenes. Dempster's normalization in (4) renormalizes conflict away. As a result, two confident but contradictory member opinions fuse to a deceptively low vacuity (Zadeh's paradox). Such contradictory opinions are the signature of an object that moved between mapping passes. We therefore accumulate the mean pairwise conflict $\bar{\eta}_i$ alongside the fused mass. We use the effective vacuity $\tilde{u}_i = \min(1,\, u_i + \bar{\eta}_i)$ in all downstream weighting. An equivalent option is Yager's variant [28] of (4). That variant assigns the conflict mass to vacuity instead of renormalizing. With this correction the primitive vacuity reflects both lack of evidence and persistent disagreement, rather than single-frame noise. The map is the set $\mathcal{M} = \{\mathcal{G}_i\}$ of (2). For retrieval, it is partitioned into overlapping submaps sampled along the mapping trajectory [2].

B. Evidential Query Encoding

Given I_q, we obtain instance masks with Fast-SAM [24] and crop each instance. Each crop passes through CLIP [23] to obtain f_j^q. It also passes through the same evidential head as the map to obtain $(\mathbf{p}_j^q, u_j^q)$ via (3). Per-instance geometry (normalized centroid $\mathbf{x}_j \in [0,1]^2$ and area a_j) is encoded by small MLPs. A query instance descriptor is

$$d_j^q = \mathrm{MLP}\big[\, \phi(f_j^q) \,\|\, \psi_{\mathbf{x}}(\mathbf{x}_j) \,\|\, \psi_a(a_j) \,\big], \tag{5}$$

with ϕ a projection of the CLIP feature into the shared space and $\|$ concatenation. Reference descriptors d_i^G are formed analogously from $(f_i, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ projected into the camera frame. Because both modalities pass through the same evidential head, u^q and u^G are directly comparable.

C. Uncertainty-Weighted Matching and the Pose Mixture

Gated instance similarity. We score a cross-modal instance pair by appearance similarity modulated by an evidential gate. The gate suppresses pairs in which either side is uncertain:

$$s_{ij} = \underbrace{\cos\big(d_i^G,\, d_j^q\big)}_{\text{semantic similarity}} \cdot \underbrace{\exp\big(-\lambda_u\,(u_i^G + u_j^q)\big)}_{\text{evidential gate}}, \tag{6}$$

where $\lambda_u \geq 0$ controls gate sharpness. This is the localization analogue of the uncertainty-aware kernel in E^2-BKI [19]. Confident–confident matches dominate. High-vacuity instances contribute only weakly.

Scene descriptors and retrieval. We follow the contrastive retrieval design of SGLoc [2]. Instance descriptors are aggregated into a scene-level descriptor by multi-head self-attention with a feed-forward layer. From $\hat{d} = Q + \mathrm{Attn}(Q, K, V)$ and $\hat{D} = \hat{d} + \mathrm{FFN}(\hat{d})$, we obtain attention weights $W = \mathrm{softmax}(\mathrm{MLP}(\hat{D}))$ and the scene descriptor $F = \sum_j \hat{D}_j W_j$ for each modality. Cosine similarity between the query scene descriptor F^q and each submap descriptor F_k^G ranks submaps. The top-k submaps provide k coarse pose hypotheses with retrieval-score weights $\tilde{\pi}_k \propto \mathrm{sim}(F^q, F_k^G)$. The gate (6) is injected into the instance-level term of the scene aggregation. Retrieval therefore inherits the same uncertainty weighting. Ambiguous scenes return several comparably scored submaps. The hypotheses therefore form a pose mixture. This mixture is the $SE(3)$ realization of the non-Gaussian error observed in [14].

Training objective. The retrieval encoders are trained with a symmetric InfoNCE loss that aligns 2D and 3D scene descriptors [2]. The semantic heads are trained with the evidential loss of [20]: the Bayes risk of the Brier score under the Dirichlet plus a KL term that drives spurious evidence toward the uniform prior,

$$\mathcal{L}_{\mathrm{EDL}} = \sum_c \Big[(y_c - \hat{p}_c)^2 + \tfrac{\hat{p}_c(1-\hat{p}_c)}{S+1}\Big] + \lambda_t\, \mathrm{KL}\big[D(\tilde{\boldsymbol{\alpha}}) \,\|\, D(\mathbf{1})\big], \tag{7}$$

with $\tilde{\boldsymbol{\alpha}}$ the evidence of misleading classes. Refinement (below) is training-free.

D. Uncertainty-Aware Refinement and Pose Covariance

Weighted rendering objective. Each coarse hypothesis T_0 is refined by differentiable rendering [10]. We parameterize the update in the tangent space as $T(\boldsymbol{\xi}) = \exp(\boldsymbol{\xi}^\wedge)\, T_0$ with $\boldsymbol{\xi} \in \mathfrak{se}(3)$. We render both an RGB image $I_r(\boldsymbol{\xi})$ and a per-pixel uncertainty map U_r. The map U_r is obtained by alpha-blending primitive vacuity exactly as color/depth are blended in 3DGS:

$$U_r(\mathbf{x}) = \sum_i u_i\, \alpha_i(\mathbf{x}) \prod_{j<i} \big(1 - \alpha_j(\mathbf{x})\big), \tag{8}$$

where $\alpha_i(\mathbf{x})$ is the opacity contribution of primitive i at pixel $\mathbf{x}$. The refinement minimizes a per-pixel weighted photometric term plus a sparse matching term [5]:

$$E(\boldsymbol{\xi}) = \sum_{\mathbf{x}} w(\mathbf{x})\, \big\| I_q(\mathbf{x}) - I_r(\mathbf{x}; \boldsymbol{\xi}) \big\|^2 + \lambda \sum_k \big\| \mathbf{x}_k^q - \mathbf{x}_k^r \big\|^2, \tag{9}$$

with $w(\mathbf{x}) = \exp\big(-\gamma\, U_r(\mathbf{x})\big)$. Dynamic and ambiguous pixels receive small weight continuously. Such pixels include movable objects and texture-poor or semantically uncertain regions. This graded, learned-uncertainty-driven weighting replaces the binary robust mask of [3]. The weights are held fixed at each iterate (IRLS-style). Gradients flow through the photometric residual only. If w were differentiated through, the optimizer could reduce (9) by steering high-residual pixels into high-vacuity regions instead of correcting the pose.

Covariance via Laplace approximation. At the optimizer's solution $\boldsymbol{\xi}^\star$, let $\mathbf{r}$ be the stacked weighted residuals of (9) and $J = \partial \mathbf{r}/\partial \boldsymbol{\xi}|_{\boldsymbol{\xi}^\star}$ the Jacobian. A Laplace approximation of the per-hypothesis posterior is a tangent-space Gaussian

$$\boldsymbol{\xi} \sim \mathcal{N}\big(\boldsymbol{\xi}^\star,\ \boldsymbol{\Sigma}_{\boldsymbol{\xi}}\big), \qquad \boldsymbol{\Sigma}_{\boldsymbol{\xi}} \approx \hat{\sigma}^2 \left(J^\top W J\right)^{-1}, \tag{10}$$

where $W = \mathrm{diag}\{w(\mathbf{x})\}$ and $\hat{\sigma}^2$ is estimated from the residuals. A flat optimum (poorly constrained pose) yields a large covariance. A sharp basin yields a tight one. The reported uncertainty therefore reflects the actual conditioning of the alignment. Real photometric residuals are heavy-tailed and spatially correlated. The raw Laplace covariance can therefore remain optimistic. We apply a single scalar inflation factor calibrated on a held-out split (a temperature on $\boldsymbol{\Sigma}_{\boldsymbol{\xi}}$). We ablate raw vs. tempered covariances in Sec. V.

Pose distribution and condensation. A PSNR/feature-similarity check [2] prunes hypotheses whose renders diverge from I_q. The surviving hypotheses form a Gaussian mixture on $SE(3)$, with each component expressed in the tangent space at its mode:

$$p\big(T \,|\, I_q, \mathcal{M}\big) = \sum_k \pi_k\, \mathcal{N}\big(\boldsymbol{\xi};\ \boldsymbol{\xi}_k^\star,\ \boldsymbol{\Sigma}_{\boldsymbol{\xi},k}\big), \tag{11}$$

with weights π_k renormalized from $\tilde{\pi}_k$ and the post-refinement render fidelity. For a downstream Bayesian filter, we condense (11) to a single Gaussian by moment matching [14]: $\boldsymbol{\mu} = \sum_k \pi_k \boldsymbol{\xi}_k^\star$ and $\boldsymbol{\Sigma} = \sum_k \pi_k \big(\boldsymbol{\Sigma}_{\boldsymbol{\xi},k} + (\boldsymbol{\xi}_k^\star - \boldsymbol{\mu})(\boldsymbol{\xi}_k^\star - \boldsymbol{\mu})^\top\big)$. This Gaussian is transported to the inertial frame as the measurement covariance for an EKF/UKF [4]. As an alternative for the rotational part, the orientation may be modeled by a Bingham distribution on the unit quaternion. We evaluate both options in the ablations.

Algorithm 1 USGLoc inference (one query image)

Require: query I_q; evidential map $\mathcal{M} = \{\mathcal{G}_i\}$; submaps $\{S_k\}$
1: $\{\mathcal{Q}_j\} \leftarrow$ Fast-SAM masks of I_q; CLIP + EDL per instance (3)
2: build query descriptors d_j^q (5) and scene descriptor F^q
3: score submaps by gated similarity (6); take top-k $\rightarrow$ hypotheses $\{(T_0^k, \tilde{\pi}_k)\}$
4: for each hypothesis k do
5: refine T_0^k by minimizing $E(\boldsymbol{\xi})$ (9) with weight $w = e^{-\gamma U_r}$ (8)
6: compute $\boldsymbol{\Sigma}_{\boldsymbol{\xi},k}$ by Laplace approximation (10)
7: end for
8: prune by render fidelity; assemble GMM (11); condense for the filter
9: return $p(T \,|\, I_q, \mathcal{M})$ and its single-Gaussian summary

V. Experiments

A. Experimental Setup

Datasets. We evaluate on three public indoor benchmarks of increasing scale and difficulty: 7Scenes [29], 12Scenes [30], and ScanNet [31]. We also evaluate on a semi-dynamic dataset captured with object rearrangements between mapping and query. This follows the protocol of [3]. On 7Scenes we adopt the SfM pseudo-ground truth [32]. This is the convention used by all recent 3DGS localizers [12], [13], [16], [26]. DSLAM-referenced numbers are not directly comparable and are discussed separately. One example is SGLoc [2], which also evaluates only four of the seven scenes. [TODO: optionally add your own indoor-MAV sequences.]

Baselines. APR: PoseNet [6], Marepo [7]. SCR: ACE [9], DSAC* [8]. 3DGS-based: iComMa [10], GS-CPR [12], GSplatLoc [11], STDLoc [13], the uncertainty-guided point estimator UGS-Loc [16], and the semantic 3DGS localizers SGLoc [2] and Chu–Kim [3]. SplatHLoc [26] is concurrent work. For calibration, no prior 3DGS localizer reports a covariance. We therefore compare against a softmax (non-evidential) variant of our method, point baselines wrapped in the best fixed covariance, and the ideal χ^2 reference.

Metrics. Accuracy: median translation (cm) and rotation (°) error, and success rate at 5 cm/5° and 2 cm/2°. Calibration is our focus. We report four metrics. (i) Coverage at the 68.3/95.4/99.7% credibility levels. A query is covered when the Mahalanobis error $d^2 = \mathbf{e}^\top \boldsymbol{\Sigma}^{-1} \mathbf{e}$ falls inside the corresponding χ^2 quantile. We compare against the ideal χ^2 curve as in [14]. (ii) AUSE (sparsification error) [33] for translation and rotation. This tests whether ranking predictions by predicted uncertainty and discarding the worst monotonically reduces error. (iii) Negative log-likelihood (NLL) of the ground-truth pose under (11). (iv) Expected calibration error (ECE) [34]. Runtime is reported in Hz.

Implementation. The map uses an open-vocabulary semantic 3DGS [27]. Submaps are cubic regions sampled along the trajectory. The EDL segmentation head and retrieval encoders are trained with Adam ([TODO: lr, epochs, batch]). Refinement is training-free with N=[$TODO$:] iterations. The parameters λ, γ in (9) are set by grid search. The covariance temperature (Sec. IV-D) is fitted once per scene on a held-out calibration split. This split is disjoint from all test queries. We use the robust median estimator $\tau = \mathrm{med}(d^2)/\chi^2_{6,0.5}$. We develop on an RTX 4090. We report embedded throughput on a Jetson Orin AGX. [TODO: state k, λ_u, C/vocabulary, segmentation backbone.]

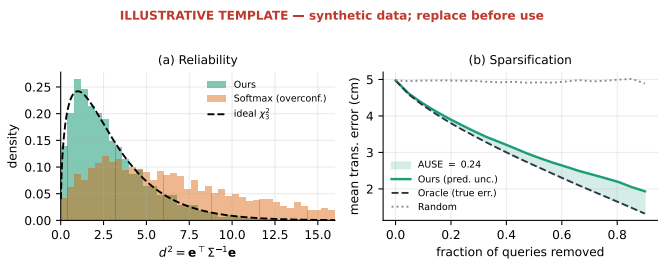


Fig. 3. Calibration diagnostics. (a) Reliability: the normalized squared error $d^2 = \mathbf{e}^\top \boldsymbol{\Sigma}^{-1} \mathbf{e}$ follows χ^2 when the covariance is calibrated. An overconfident softmax variant exhibits a heavy tail. (b) Sparsification: ranking queries by predicted uncertainty (Ours) should track the oracle (rank by true error). AUSE is the area between them. Template with synthetic data. Regenerate with measured results using fig3_calibration_template.py.

B. Localization Accuracy

Table I reports median errors on 7Scenes. The strongest published point estimators are SplatHLoc [26] (0.55 cm/0.17°) and the uncertainty-guided UGS-Loc [16] (0.64 cm/0.21°, ≈2.2 s/query). Our goal is to remain competitive with this group. We additionally produce a calibrated covariance that none of them reports. [TODO: 1–2 sentences with measured numbers.] SGLoc [2] reports 0.5 cm/0.15° prior-free. However, it uses an easier protocol: 4/7 scenes under DSLAM ground truth, omitting Fire, Pumpkin, and Stairs. Its code is unreleased. We therefore compare against its protocol in the supplementary. [TODO: decide whether to keep the SGLoc-protocol comparison.]

C. Uncertainty Calibration

This is the central evaluation. Prior 3DGS localizers report no calibration numbers. We therefore compare against an ablated softmax variant of our own method and against the ideal χ^2 reference. Fig. 3 and Table II summarize coverage, AUSE, NLL, and ECE. [TODO: state that coverage tracks the nominal credibility levels and that AUSE is low (uncertainty is informative).]

D. Robustness

We stress three axes with established protocols. (i) Low-quality maps: maps built from ∼300 vs. ∼1000 RGB images [3]. (ii) Input sparsity: mapping from a reduced fraction of frames [19]. (iii) Degraded semantics: training the segmentation head on progressively less data [19]. [TODO: report that USGLoc degrades more gracefully and, crucially, that its uncertainty grows appropriately as conditions worsen. The covariance should be honest, not merely small.]

E. Ablation Study

Table III isolates each component. [TODO: summarize: EDL vs softmax semantics; with/without the evidential gate (6); with/without per-pixel uncertainty weighting (9); Dempster vs. conflict-aware fusion (effective vacuity / Yager); point estimate vs. mixture (K=1 vs. K=3); raw vs. tempered Laplace covariance; tangent-Gaussian vs. Bingham rotation.]

F. Runtime

[TODO: report per-stage timing on RTX 4090 and Jetson Orin AGX; note that the Laplace covariance reuses the refinement Jacobian and adds negligible overhead; contextualize against UGS-Loc's reported ≈2.2 s/query (8 particles, 2 iterations) [16].]

VI. Conclusion

We presented USGLoc, a semantic 3D Gaussian Splatting localizer that returns a calibrated distribution over camera pose rather than a point. USGLoc is built on one principle: the map's own uncertainty drives every stage of localization and survives to the output. A single evidential state is created when observations are encoded, and it is never discarded. It gates 2D–3D correspondences during matching. It weights the rendering-refinement residual per pixel. It scales the Laplace covariance through a held-out temperature. These steps propagate uncertainty end-to-end into a Gaussian-mixture posterior on $SE(3)$. The posterior condenses to a single Gaussian for Bayesian filtering. [TODO: one closing sentence on headline results.] Future work includes a fully open-set semantic extension, tighter coupling with the navigation filter, learned (rather than Laplace) covariance heads, and on-board flight experiments on an indoor MAV.

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TABLE I
Median translation/rotation error (cm/°) on 7Scenes under the SfM pseudo-ground truth [32]. Lower is better. Rows marked † are pre-filled as reported in the cited papers. GS-CPR and UGS-Loc refine an ACE prior. PoseNet/Marepo rows are the SfM re-evaluations of [12]. iComMa [10] publishes no 7Scenes results and is compared in the ablations. SGLoc [2] evaluates only 4/7 scenes under DSLAM ground truth and is discussed in the text. Ours: placeholders.

Method	Chess	Fire	Heads	Office	Pumpkin	Kitchen	Stairs	Avg.
PoseNet [6]†	10/4.02	27/10.0	18/13.0	17/5.97	19/4.67	22/5.91	35/10.5	21/7.74
Marepo [7]†	1.9/0.83	2.3/0.92	2.1/1.24	2.9/0.93	2.5/0.88	2.9/0.98	5.9/1.48	2.9/1.04
DSAC* [8]†	0.5/0.17	0.8/0.28	0.5/0.34	1.2/0.34	1.2/0.28	0.7/0.21	2.7/0.78	1.1/0.34
ACE [9]†	0.5/0.18	0.8/0.33	0.5/0.33	1.0/0.29	1.0/0.22	0.8/0.20	2.9/0.81	1.1/0.34
GSplatLoc [11]†	0.4/0.13	0.9/0.29	0.9/0.50	1.4/0.32	1.4/0.26	1.3/0.29	3.4/0.82	1.4/0.37
GS-CPR [12]†	0.5/0.15	0.6/0.25	0.4/0.28	0.9/0.26	1.0/0.23	0.7/0.17	1.4/0.42	0.8/0.25
STDLoc [13]†	0.5/0.15	0.6/0.24	0.5/0.26	0.9/0.24	0.9/0.21	0.6/0.19	1.4/0.41	0.8/0.24
UGS-Loc [16]†	0.4/0.12	0.5/0.20	0.4/0.25	0.8/0.22	0.8/0.18	0.6/0.15	1.1/0.33	0.6/0.21
SplatHLoc [26]†	0.3/0.09	0.4/0.19	0.3/0.18	0.6/0.18	0.7/0.15	0.5/0.12	1.0/0.30	0.6/0.17
USGLoc (Ours)	–	–	–	–	–	–	–	–

TABLE II
Calibration on ScanNet. Coverage should match the nominal level; AUSE/NLL/ECE lower is better. Placeholders.

Method	Cov. 68 (%)	Cov. 95 (%)	Cov. 99.7 (%)	AUSE↓ (t/R)	NLL↓
Ideal χ^2 ref.	68.3	95.4	99.7	–	–
Softmax variant	–	–	–	–	–
USGLoc (Ours)	–	–	–	–	–

TABLE III
Ablation on ScanNet. Placeholders.

Variant	Med. err.↓	SR@5/5↑	AUSE↓	NLL↓
Full USGLoc	–	–	–	–
− gate (6)	–	–	–	–
− pixel weight (9)	–	–	–	–
Dempster (no conflict)	–	–	–	–
softmax (no EDL)	–	–	–	–
point estimate (K=1)	–	–	–	–
raw Laplace (no τ)	–	–	–	–

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